



TED UNIVERSITY



Introduction

The flight safety, fuel efficiency, and overall performance of an aircraft are directly dependent on the precise location of the Center of Gravity (CG). Current methods often rely on static pre-flight measurements or manual data entry, which are time-consuming and prone to human error. This project introduces a fully integrated embedded system that provides real-time, dynamic CG detection and autonomous balancing.



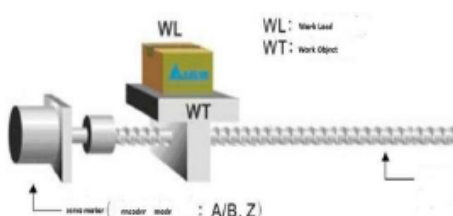
Data Acquisition & Sensing Unit

This unit captures real-time force data via high-precision load cells. The 24-bit HX711 ADC digitizes these signals to provide high-resolution input for precise Center of Gravity (CG) calculations and autonomous balancing.

CHALLENGES

- Software filters to eliminate electrical noise in load cell data.
- Minimizing response delay between sensor detection and actuator displacement.
- Ensuring measurement precision under varying environmental conditions.
- Overcoming counterweight inertia to prevent system oscillations.

BALANCING SYSTEM



Center of Gravity (CG) Detection, Notification, and Balancing System for Aircraft

Technical Goals & Development Objectives

- ✓ **Real-Time CG Monitoring:** Designing an embedded architecture to calculate weight distribution dynamically.
- ✓ **Autonomous Equilibrium Control:** Engineering a mechanical unit for millimetric CG correction via closed-loop feedback.
- ✓ **Precision Data Acquisition:** Processing 24-bit digital signals from load cells for high-fidelity force detection.
- ✓ **Aviation Process Digitalization:** Reducing flight trial cycles by providing high-accuracy pre-flight calibration.
- ✓ **System Stability Optimization:** Implementing control logic to manage weight shifts without mechanical oscillation.

METHOD

Design Approach: A multidisciplinary engineering framework integrating physical load analysis, embedded software development, and autonomous mechanical correction.

System Components: Arduino Uno (ATmega328P), 24-bit HX711 ADC modules, Full-bridge Load Cell sensors, and dual-axis lead screw actuators.

Software Design Principles:

- Real-time sensor-to-actuator control loop for instant stabilization.
- Automated calculation of CG_x and CG_y coordinates relative to a fixed datum.
- High-frequency 24-bit digital signal handling for millimetric accuracy.
- Real-time derivation of counterweight displacement to neutralize detected deviations.

Initial Product Demonstration:

- **Sensing:** Three full-bridge load cells positioned under the landing gear.
- **Balancing:** A dual-axis lead screw assembly driven by 360° servo motors.
- **Correction:** Automatic displacement calculation based on detected deviations

Functional Development & System Integration

A. Prototype Evaluation & Balancing

The system uses a 3D helicopter prototype to determine the initial Center of Gravity (CG). When external weight is added, the sensing platform detects force changes, and the dual-axis lead screw mechanism repositions the counterweight to restore equilibrium.

B. Real-Time Data Processing

Analog load cell signals are digitized via HX711 and processed by an Arduino-based controller. The algorithm performs real-time moment calculations to update X-Y coordinates and drives servo-actuators with millimetric precision.

C. UI Design & Monitoring Interface

A custom User Interface (UI) provides live visualization of stability status and total added weight. A dynamic coordinate system tracks the CG shift instantly, allowing operators to monitor the autonomous balancing process through serial communication.

Conclusion

This project delivers a fully autonomous "detect-and-correct" platform, transforming aircraft weight management from manual procedures into a high-precision digital process, enhancing safety and efficiency for the domestic aviation industry.

